

Danila Arinov

C++ / Rust Developer — Graphics, Engines & Tools

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PROFILE

C++/Rust developer focused on low-level graphics and tools: engines, editors, rendering. Hands-on practice since 2024: open-source, team development, paid DnD art. Flagship project — **TPaint**: a Rust/OpenGL raster editor with a custom UI, 15+ tools and 140+ automated tests. Team Unity game about the Middle Ages (code, game design, hiring) plus paid artist work on a DnD project (all visuals).

EXPERIENCE

TPaint — Raster Editor (SAI / Clip Studio analogue)

Rust · OpenGL / GLFW · Aug 2026 – Present

- Designed and built a custom immediate-mode UI in OpenGL from scratch: panels, color picker, file dialogs — zero native controls.
- Implemented 15+ tools (brush, eraser, shapes, fill, Cyrillic text): symmetry up to 12 sectors, stroke stabilizer, sticky and healing brushes.
- Implemented layers: folders, masks, 10 blend modes, effects (shadow/outline/glow), filters (curves, levels, hue/balance) — each edit is one undo step.
- Implemented formats: custom .tpaint, PSD import/export with layers, PNG/JPEG; per-minute autosave, undo/redo with a clickable history strip.
- Covered with 140+ automated tests (raster, layers, history, PSD, UI logic); MSVC + MinGW builds, dark theme, DPI-aware. First stable release published on GitHub.

github.com/TeivrimOriginal/Copy-SAI-Paint-with-Rust

Teivrim-Engine — Game Engine from Scratch

C++17 · Vulkan / OpenGL · CMake · Oct 2025 – Present

- Designed a modular architecture: runtime backend selection via an AbstractRender singleton, one scene codebase for Vulkan and OpenGL.
- Implemented a scene graph (parent/child, quaternions, world matrices), FBX/OBJ/MTL import via Assimp, text via stb_truetype atlases.
- Set up GLSL → SPIR-V shaders, debugging with Khronos validation layers, CMake build with vendored deps; 88 commits.

github.com/TeivrimOriginal/Teivrim-Engine

Anime DB — Catalogue from 3 Public APIs

Rust · Actix-Web · SQLite · Docker · 2024 – Present

- Rewrote the first version from Node.js in Rust and improved the data-extraction algorithm for public APIs.
- Aggregated AniList (~20k records) + Kitsu + Shikimori into one SQLite DB with FTS5 (unicode61 — Russian search works).
- Implemented a same-origin JSON API: filters, pagination, suggest, rate limiting, static mobile-first frontend (RU/EN) plus a native Android client.
- Implemented accounts: Argon2id, opaque tokens (SHA-256, 90 days); hand-rolled mini connection pool and CORS to keep dependencies lean.

github.com/TeivrimOriginal/TeivrimSite

Visual Novel Editor

C++ · Win32 / GDI+ / MinGW

- Developed 6 panels: Hierarchy, Scene, Inspector, Script, Asset Browser, Timeline; drag-and-drop from Explorer, project save/load, tick-based dialogue preview.

github.com/TeivrimOriginal/teivrim-novell-engine

TEAM EXPERIENCE

- Medieval Unity indie game — code, game design, hiring people for the project; preparing the project for investor outreach.
- Artist on a private DnD project (paid) — all visuals: characters, environments, assets; one style for DMs and players.

- HackIT hackathon — graphics part and build; shipped a demo prototype over one weekend.

SKILLS

Languages: C++17, Rust, Python, GLSL, SQL (SQLite)

Graphics: OpenGL, Vulkan, Win32 API, GDI/GDI+, GLFW, GLEW, GLM, Assimp, stb

Backend / Tools: Actix-Web, SQLite FTS5, CMake, Cargo, Git, Docker, MinGW / MSVC

Practices: testing, review, SPIR-V, architecture docs

ADDITIONAL

English — reading technical documentation. Previously — Qt 6 (~2 years, 2–3 projects), now focused on Rust/C++. Past freelance on Kwork (orders via kwork.ru/user/teivrim). Understand the art pipeline from sketch to asset (Blender, 2D).